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Section: F

Problem Statement: *Daily Horoscope Generator*

Abstract:

This project implements a Daily Horoscope Generator using a pre-trained Large Language Model (LLM) from the Hugging Face Transformers library. The system takes a zodiac sign as input and generates a short, meaningful horoscope in natural language. Instead of training a model from scratch, a text generation pipeline is used to demonstrate how large language models can be applied to real-world natural language generation tasks. The project highlights the practical use of transformer-based models for creative text generation and provides an easy-to-understand implementation suitable for beginners in Generative AI.

Documentation:

This project implements a Daily Horoscope Generator using a pre-trained Large Language Model from the Hugging Face Transformers library. The system takes a zodiac sign as input and generates a short horoscope using a text generation pipeline. No model training or dataset is required, as the project relies on an existing transformer-based model. The implementation demonstrates how Large Language Models can be used for natural language generation tasks in a simple and practical manner.

OUTPUT

```
zodiac_dropdown = widgets.Dropdown(
    options=[
        "Aries", "Taurus", "Gemini", "Cancer",
        "Leo", "Virgo", "Libra", "Scorpio",
        "Sagittarius", "Capricorn", "Aquarius", "Pisces"
    ],
    description="Zodiac:"
)

display(zodiac_dropdown)
```

Zodiac:

```
generate_button = widgets.Button(description="Generate Horoscope")

def on_button_click(b):
    generate_horoscope(zodiac_dropdown.value)

generate_button.on_click(on_button_click)
display(generate_button)
```

Generate Horoscope

Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate examples to max length. Defaulting to 'long'. Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation. Both 'max_new_tokens' (~250) and 'max_length' (~60) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main_classes/text_generation)

Today's horoscope for Leo:

3/18/18 - The Moon is the largest star of the year and Leo is considered a major star in the sky.

- The Moon is the largest star of the year and Leo is considered a major star in the sky. 3/19/18 - The Year of the Emperor and Leo is the most infamous of all the years.
- The Year of the Emperor and Leo is the most infamous of all the years. 3/22/18 - Leo is the most popular celebrity worldwide and is considered a star of popular life with the average person.
- Leo is the most popular celebrity worldwide and is considered a star of popular life with the average person. Leo is also considered to be the most important star of the planet and
- The world's largest star is known as the Milky Way. 2/18/18 - Leo is known to be a major star of astronomy because of its shape. He is believed to be the smallest star in the sky, 4

1. What is the objective of this project?

Answer:

The objective is to generate daily horoscopes for zodiac signs using a pre-trained Large Language Model without training a model from scratch.

2. Which model or library is used in this project?

Answer:

The project uses the Hugging Face Transformers library with a pre-trained GPT-based text generation model.

3. Why did you use a pre-trained model instead of training your own?

Answer:

Pre-trained models save time and computational resources and are suitable for demonstrating natural language generation tasks in Unit-1.

4. What is the role of the Hugging Face pipeline?

Answer:

The pipeline simplifies the process of loading the model and generating text with minimal code.

5. Is any dataset used or model trained in this project?

Answer:

No, the project does not use any dataset or perform model training.

6. What type of Generative AI task does this project demonstrate?

Answer:

It demonstrates a text generation task using a Large Language Model.

7. What are the key learnings from this project?

Answer:

Understanding how to use pre-trained transformer models and Hugging Face pipelines for practical text generation applications.